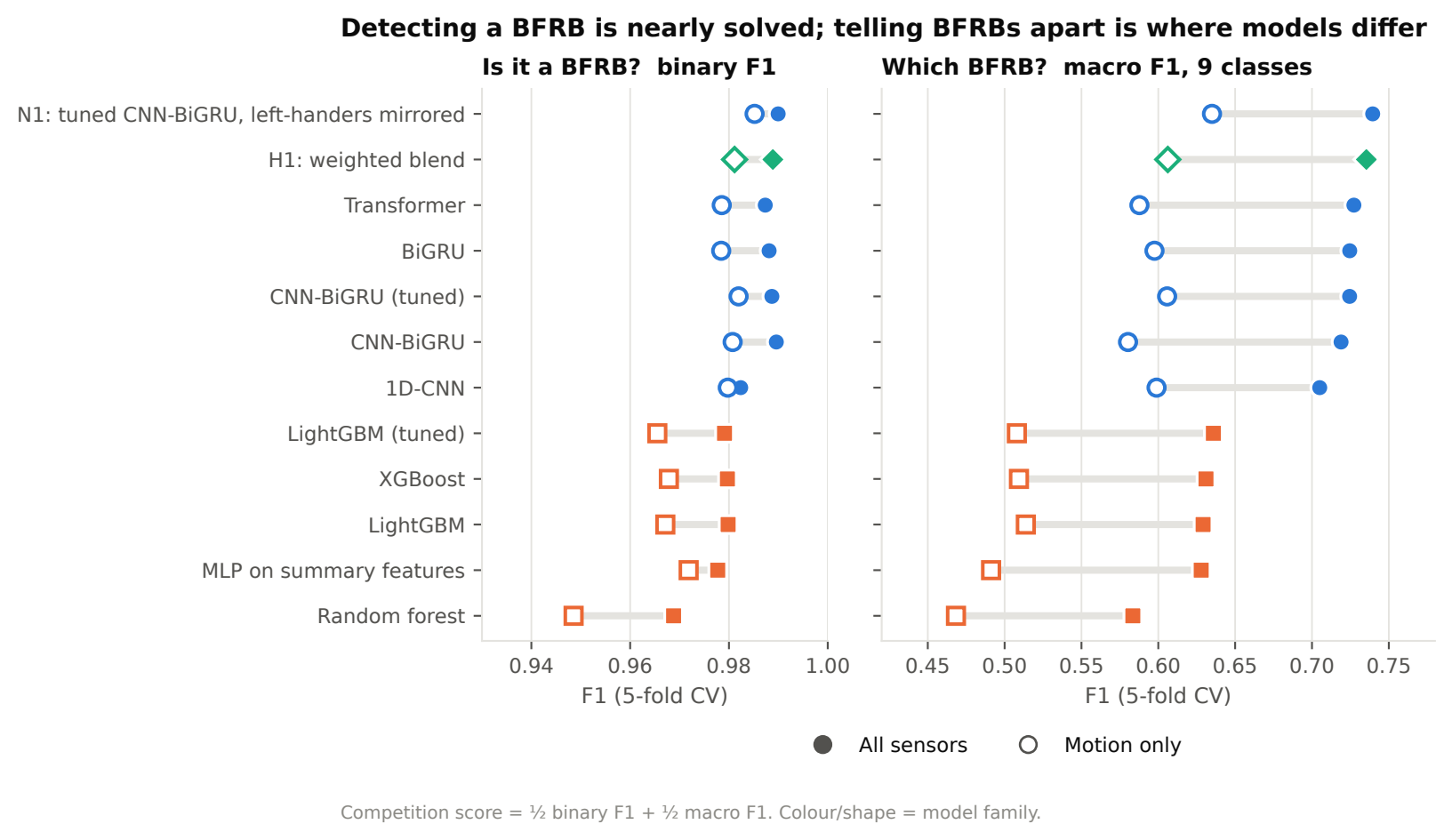
Detecting a BFRB is nearly solved; telling BFRBs apart is where models differ
Is it a BFRB? binary F1
Which BFRB? macro F1, 9 classes
N1: tuned CNN-BiGRU, left-handers mirrored
H1: weighted blend
Transformer
BiGRU
CNN-BiGRU (tuned)
CNN-BiGRU
1D-CNN
LightGBM (tuned)
XGBoost
LightGBM
MLP on summary features
Random forest
0.94
0.96
0.98
1.00
F1 (5-fold CV)
0.45
0.50
0.55
0.60
0.65
0.70
0.75
F1 (5-fold CV)
All sensors
Motion only
Competition score = ½ binary F1 + ½ macro F1. Colour/shape = model family.